MAX LEONG RUI SHENG

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EDUCATION

Singapore Management University (SMU) Expected 2029
Bachelor of Science in Computing (Computer Science)

TECHNICAL SKILLS

Languages: Python, TypeScript, JavaScript, C, C++
Frontend & Product: React, Next.js, Electron, Vite
Backend & Data: FastAPI, Supabase, Redis, Celery, SQLite, WebSockets
AI & Computer Vision: MediaPipe, OpenCV, Librosa, Whisper
Hardware: ESP32, Arduino, Blynk

CERTIFICATIONS

Google AI Professional Certificate — Coursera, Jun 2026 Verify

HACKATHON ACHIEVEMENTS

it'sPEAK (Presentation Coach) — 2nd Place, SMU .Hack Enrichment Application Programme (HEAP) 2026 | GitHub | *Python, FastAPI, Next.js, MediaPipe, Librosa, Whisper, Celery, Redis, Supabase*

- Built a private presentation-practice coach that turns a short recording into evidence-based feedback on pacing, pitch variation, filler words, pause quality, eye-contact proxy, posture, and gesture range
- Implemented a baseline-comparison system letting users measure rehearsal progress against a protected personal benchmark
- Designed an asynchronous processing pipeline (Celery + Redis) to handle audio/video analysis without blocking the user-facing app

Echo — Best Freshmen Award, NTU iNTUition 2026 | GitHub | *TypeScript, Next.js, Supabase, Computer Vision*

- Developed a reminiscence-therapy companion for people with dementia, prioritising familiar memories over novel content to reduce cognitive overload
- Designed adaptive interaction logic that adjusts pacing and content during detected moments of user vulnerability

PERSONAL PROJECTS

Software

Network Sniffer | GitHub | *Python, Scapy, FastAPI, WebSockets, Next.js*

- Built a real-time packet capture and telemetry dashboard, streaming live network traffic data from a Scapy backend to a Next.js frontend over WebSockets
- Architected a FastAPI service layer to process and relay packet-level data with minimal latency

Hologram AR | GitHub | *Python, MediaPipe, Computer Vision*

- Created a real-time webcam AR system using MediaPipe hand tracking to drive interactive overlays and gesture-triggered visual effects
- Engineered spatial interaction logic mapping tracked hand landmarks to on-screen holographic elements

Hardware

Conway's Game of Life Data Factory | GitHub | *ESP32, React, Jupyter Notebook*

- Implemented Conway's Game of Life on a 32×8 LED matrix driven by an ESP32, streaming live entropy and birth/death telemetry to a React dashboard

ESP32 Asteroids | GitHub | *ESP32, C++, OLED, PWM*

- Built a breadboard-based Asteroids arcade game on an ESP32 with a 128×64 OLED display, joystick controls, and PWM-driven sound effects
- Implemented persistent high scores, adaptive difficulty, and an AI-driven attract mode

CO-CURRICULAR ACTIVITIES

SMU Legal Innovation & Technology Society — Law Associate (EXCO)

- Contribute to a cross-disciplinary CCA (School of Law) exploring the intersection of legal practice and technology